

BUSINESS DATA MANAGEMENT

CAPSTONE PROJECT FINAL TERM



Data-Driven Sales, Inventory, and Profit Optimization in Agricultural Retail Business

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Executive Summary:

This study is rooted in original data collected from collected from Krishi Khad Bhandar, it is a family-run agricultural retail store in Belhariya (Sarwan, Gaya, Bihar) supplying fertilizers, seeds, and pesticides to farmers, dealers and retailers across nearby rural areas. Over two decades the business has accumulated transaction level sales and inventory records, but these data remain underutilized. Despite its success, the store faces four key challenges: inventory management issues, challenges in understanding sales over time, challenges in region wise revenue distribution and space constraint. This study is based on sales data collected from Krishi Khad Bhandar covering the period March 2023 to December 2025

To address these challenges, we conducted a comprehensive data-driven analysis using historical sales data. The analysis included trend analysis, correlation analysis, demand stability, space constraint and regional sales performance. Trend analysis revealed that seasonal crops particularly wheat and paddy, have fluctuating but high sales, especially during specific months July, august for paddy and November , December for wheat. Fertilizer sales, especially Urea and DAP, are largely driven by seed demand, highlighting a strong interdependence between product categories. Pesticides sale are steady with pest control contributing the highest. Correlation analysis highlighted interdependencies among Fertilizers, with closely related sales pattern within categories like sulphate, urea, DAP. Column chart showing highest revenue during July, august, November, December due to seasonal crops and interdependencies of fertilizer on seeds. Pareto chart showing about region that Bhadya and Manjhiyama contributes highest in revenue.

Based on the findings, we recommend adopting automated inventory management tools to maintain optimal stock levels and reduce expiration. Temporary staffing during peak months like July, august and November, December will allow Mr. Ajesh Kumar to balance his academic responsibilities. Expanding into home delivery services and implementing targeted seasonal marketing campaigns will improve customer satisfaction and drive sales. Additionally, optimizing the storage layout to prioritize high-demands items and removing slow-moving products can address space constraints.

Detailed Explanation of Analysis Method:

1. Trend Analysis

We are using trend analysis to understand the patterns and behaviors in sales data over time, which is critical for effective trend analysis and business planning at **Krishi Khad Bhandar** store. Given the store location in Gaya and the seasonal nature of its operations, sales are influenced by factors like kharif or rabi. By analyzing trends, we can identify peak demand periods prepare inventory accordingly, avoiding stockouts or overstocking during slow-moving months. Trend analysis also helps in recognizing growth opportunities, such as inconsistent increases in demand for certain product categories Urea and DAP , allowing to make informed decisions on stocking and space allocation. Furthermore, it provides insights into recurring patterns, such as weekly or monthly variations, and help align marketing strategies like discounts or promotions, with these cycles to maximize sales.

Steps to perform:

- **Data Preprocessing:**

Organize sales data by pivot table on other sheets and categorize into fertilizer, seeds, pesticides and then filter with product type. Aggregate daily sales into monthly totals for each SKU and category

- **Visualize Trends:**

Column charts use to visualize monthly seed and fertilizers demand across different crop categories in order to observe seasonal patterns and growth trends. By plotting months on the horizontal axis and seed demand values on the vertical axis, the charts make it easy to compare fluctuations over time and identify periods of high and low demand.

- **Identify Seasonal Peaks:**

Identifying seasonal peaks by observing months where demand values rise significantly compared to surrounding months. Sudden increases in demand indicate strong seasonal requirements for specific crops and fertilizers. After identifying peaks, the analysis compares different seed categories to understand crop-specific seasonal behavior.

- **Expected Outcome:** A clear understanding of sales patterns over time will enable Mr. Ajesh Kumar to prepare for high demand periods and prevents overstocking during slow months. Trend analysis is expected to facilitate a proactive approach to inventory and sales management, improving operational efficiency and enhancing customer satisfaction through timely availability of products.

2. Coefficient of variation(CV):

The coefficient of variation analysis evaluates demand stability for fertilizers, seeds, and crops by comparing standard deviation relative to average demand. The CV chart indicates that most SKUs experience moderate to high demand variability, which directly affects inventory risk and safety stock requirements. Although inventory stock data is unavailable, inventory management issues are analyzed using sales demand patterns. High variability, seasonal fluctuations, and irregular demand trends indicate challenges in maintaining optimal inventory levels, especially for perishable seasonal products. In the absence of inventory stock data, demand variability was analyzed using the coefficient of variations. Categories with higher CV values exhibit unstable demand, making it difficult to maintain optimal inventory levels and increasing the risk of overstocking and stock shortages. Safety stock levels are positively related to demand variability; therefore, products with higher coefficients of variation require greater buffer inventory.

Steps to perform:

- **Data preparation :**

Separate data into specific SKUs of fertilizer, seeds, pesticides

Find average of quantity and standard deviation into category-wise

- **Calculate CV%:**

- Find coefficient of variation among all categories

$$CV (\%) = \left(\frac{\text{Standard deviation}}{\text{Mean}} \right) \times 100$$

- **How to interpret:**

CV Range	Interpretation	Inventory Policy
Below 70%	Stable demand	Low safety stock
70% – 100%	Moderate variability	Balanced inventory
Above 100%	Highly variable	High safety stock

Table 1: Coefficient of variation

- **Visualize result:**
 - Create bar charts to represent coefficient of variation
- **Expected Outcome:** When demand variability is combined with mean demand levels, priority items for inventory control become clear. Paddy has the highest mean demand and high variability, making it a critical inventory item requiring both availability assurance and seasonal planning. Urea also shows high average demand with strong variability, making it a key operational stock.

3. Regional Sales Performance Analysis:

Regional Sales Performance Analysis evaluates how revenue is distributed across different geographic regions to identify high-performing and low-performing markets. The objective of regional sales performance analysis is to evaluate how total revenue is distributed across different villages and market areas served by Krishi Khad Bhandar. This analysis helps identify high-performing regions, moderate contributors, and underperforming areas, enabling region-specific sales, inventory and distribution strategies.

Steps to perform:

- **Classify villages by sales or revenue:**
The first step in performing regional sales analysis is aggregating total revenue for each region. All transaction values are grouped by region, and the sum of revenue is calculated for each location. After computing regional totals, the regions are arranged in descending order based on revenue contribution.
- **Formulas Used:**
Total revenue per region = Sum of all sales values in that region

Cumulative Revenue(i) = Sum of revenue from region 1 to region i

Cumulative % = (Cumulative Revenue ÷ Total Revenue of all regions) × 100

- **Expected Outcome:** Regional sales performance analysis was conducted to evaluate the distribution of total revenue across different regions served by Krishi Khad Bhandar. The pareto chart represents revenue generated from each region, while the cumulative percentage line helps determine how much each region contributes to total revenue. The results reveal high revenue concentration in a few regions, which increases business risk.

4. Correlation Analysis:

Correlation analysis is a powerful statistical method to examine the relationships between different SKUs at Krishi Khad Bhandar store. Understanding these relationship helps uncover patterns in sales and customer buying behavior. For instance, if sales of Urea strongly correlate with DAP, it indicates that customers often buy them together, highlighting cross-selling opportunities. Conversely, a weak or negative correlation between certain products may suggest independent purchasing behavior, allowing better segmentation of inventory. Given the store's space constraints, identifying such associations ensures that complementary products are strategically placed to maximize sales and enhance customer convenience. Additionally, correlation can help identify seasonal dependencies or external factors influencing sales, aiding better forecasting and inventory decisions.

Steps to perform:

- **Calculate correlation:**
 - Use Pearson correlation formula

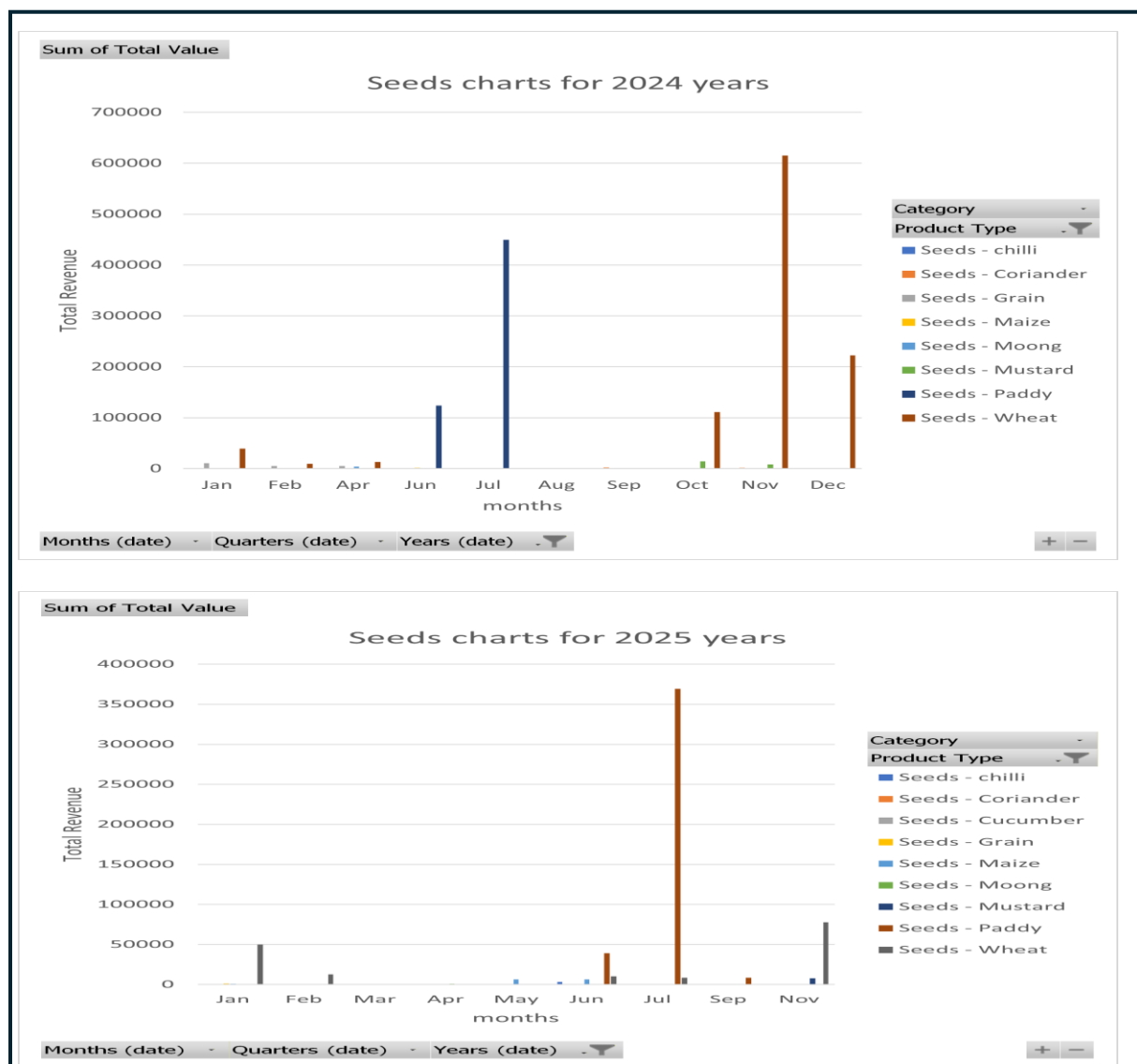
$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{\sum(X - \bar{X})^2} \sqrt{\sum(Y - \bar{Y})^2}}$$

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- **Interpret Results :**
 - Positive correlations indicate complementary products, while negative correlations suggest substitutable items.
- **Visualize relationship:**
 - Plot a heatmap for SKU categories like fertilizer, pesticides, seeds
- **Expected Outcome:** Understanding Sales dependencies across categories like Fertilizers, pesticides, Seeds will enable the store to anticipate demand shifts

and adjust inventory levels accordingly, minimizing wastages and stockouts. Correlation analysis will drive smarter inventory and marketing strategies, ultimately increasing profitability and customer satisfaction.

Results and Findings:

Based on the trend analysis and category-specific insights, we observe distinct sales pattern that provide valuable information for inventory management, sales strategies, and overall business planning for Krishi Khad Bhandar store. Trend analysis examines how sales values change over time to identify patterns, seasonality, peaks and low-demand periods.



I **Figure 1: Column charts showing seeds Total revenue-wise trends**

The seed sales data for 2024 shows strong seasonal concentration with revenue heavily driven by major crop cycles. Paddy sales dominate the mid-year period, with a sharp peak in June with revenue ₹123720 and an even higher surge in July with revenue ₹449232. This pattern reflects monsoon sowing preparation, where demand rises rapidly within a short window. Wheat sales show a distinct late-year spike, with very high revenue ₹614995 in November and continued strong performance in December, indicating winter crop planning. Mustard and moong display moderate increases during the post-monsoon period, while chilli, coriander, and maize contribute smaller but periodic sales throughout the year. Overall, 2024 demonstrates a highly seasonal structure where a few crops generate the majority of revenue during specific agricultural windows.

The 2025 seed sales pattern maintains the same seasonal structure but shows changes in demand intensity and distribution. Paddy continues to be the dominant revenue contributor, with a major peak in July with revenue ₹369267, confirming its central role in monsoon cultivation cycles. Wheat sales appear more distributed across early and late months compared to 2024, suggesting broader planting activity or supply variation. Additional crops such as cucumber appear in the sales mix, indicating product diversification. Compared to 2024, revenue peaks are fewer but still strongly concentrated around key sowing periods, reinforcing the seasonal dependency of seed demand.

Comparing both years reveals that seed demand remains driven by agricultural seasons rather than continuous consumption. Mid-year demand is dominated by monsoon crops such as paddy, while late-year demand is driven by winter crops such as wheat and mustard. The persistence of these seasonal peaks across both years confirms that inventory planning must be aligned with sowing cycles rather than uniform stocking throughout the year.

The seasonal trend analysis of fertilizer and seed sales for the years 2024 and 2025 reveals a strong dependency on agricultural cropping cycles. Sales remain relatively low during the off-season months from March to May, reflecting minimal farming activity during the summer period. A gradual increase in revenue is observed from June to September, corresponding to the Kharif cultivation season when farmers prepare land and apply fertilizers such as DAP, Urea, and Potash.

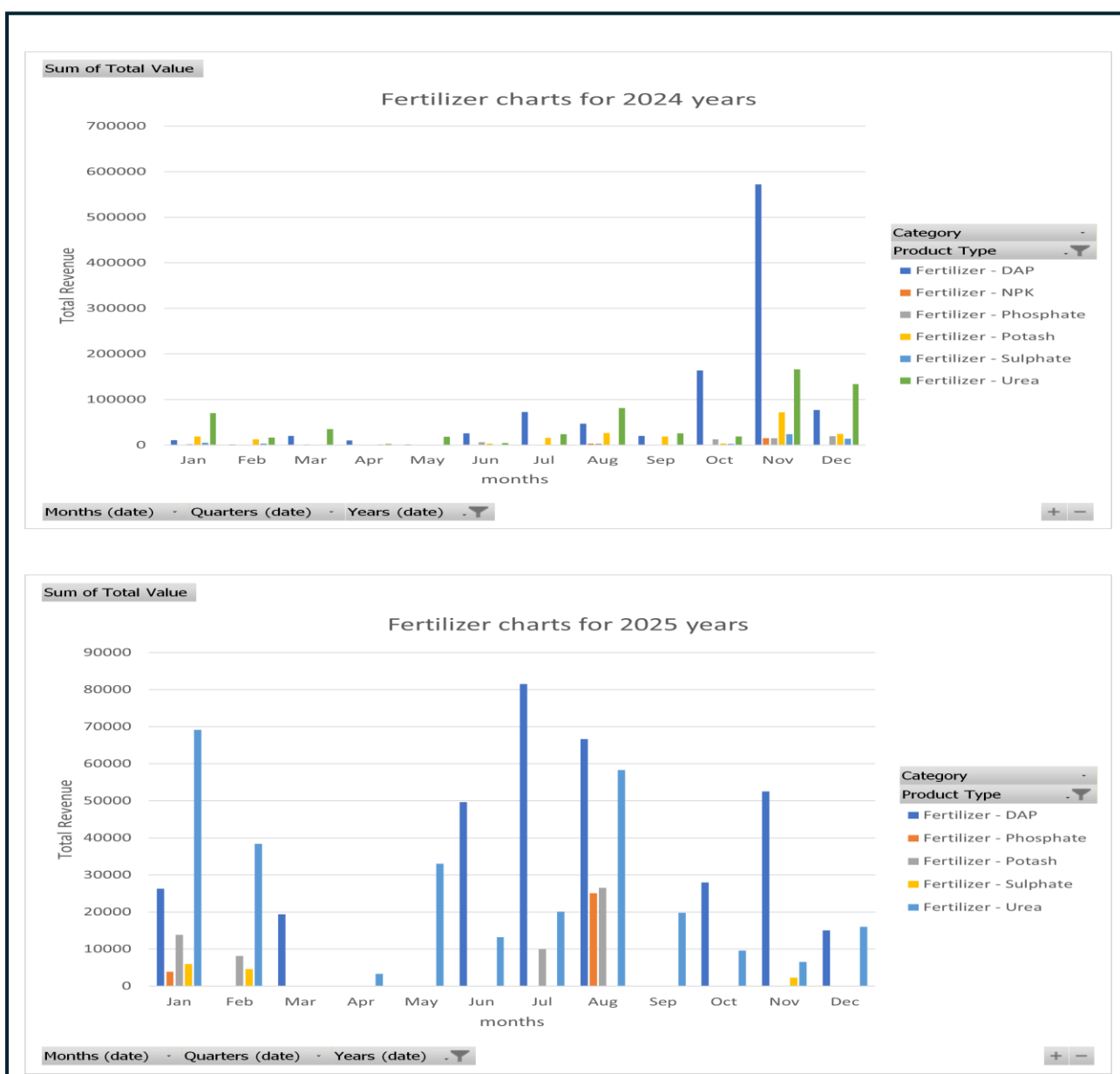


Figure 2: Column charts showing fertilizer revenue-wise trends

A significant spike in sales is recorded during October and November, particularly in 2024, indicating peak demand during the Rabi sowing season. Products such as DAP and Wheat Seeds show exceptionally high sales during these months, confirming their importance in wheat cultivation. Although 2025 also demonstrates a similar seasonal pattern, revenue distribution appears slightly more balanced compared to 2024, with notable sales increases in July, August, and November. Overall, the analysis confirms that the retail shop's revenue is highly seasonal and closely aligned with regional agricultural cycles. This pattern suggests that effective inventory planning, seasonal stocking, and cash flow management are

critical for maximizing profitability. The findings also indicate that forecasting models incorporating seasonality would be highly suitable for predicting future sales trends.

Demand stability was analyzed using the **coefficient of variation (CV)**. Products with low **CV** values were classified as stable-demand items, while high **CV** values indicated seasonal or unpredictable demand. This classification supports differentiated inventory strategies such as higher safety stock for stable items and seasonal stocking for volatile products.

SKU	Mean Demand	Std Dev	CV %	Demand Stability	Inventory Risk	Recommended Inventory Action
Urea	19.28	14.22	74%	Moderately variable	Medium	Maintain balanced safety stock
DAP	7.2	7.28	101%	Highly variable	High	Keep higher safety stock + frequent review
Sulphate	2	1.37	68%	Relatively stable	Low–Medium	Regular replenishment cycle
Potash	4.04	3.14	78%	Moderately variable	Medium	Maintain moderate buffer stock
Phosphate	5.63	3.99	71%	Moderately stable	Medium	Standard reorder policy
Wheat	8.52	8.36	98%	Highly seasonal	High	Seasonal stocking before peak demand
Paddy	24.39	18.64	76%	Predictable seasonal demand	Medium	Pre-season bulk procurement
Mustard	10.32	12	116%	Highly unstable demand	Very High	High safety stock + close monitoring
Spinach	4.86	6.27	129%	Most volatile demand	Very High	Flexible inventory + quick replenishment

Table 2: Showing demand stability and inventory risk using coefficient of variance

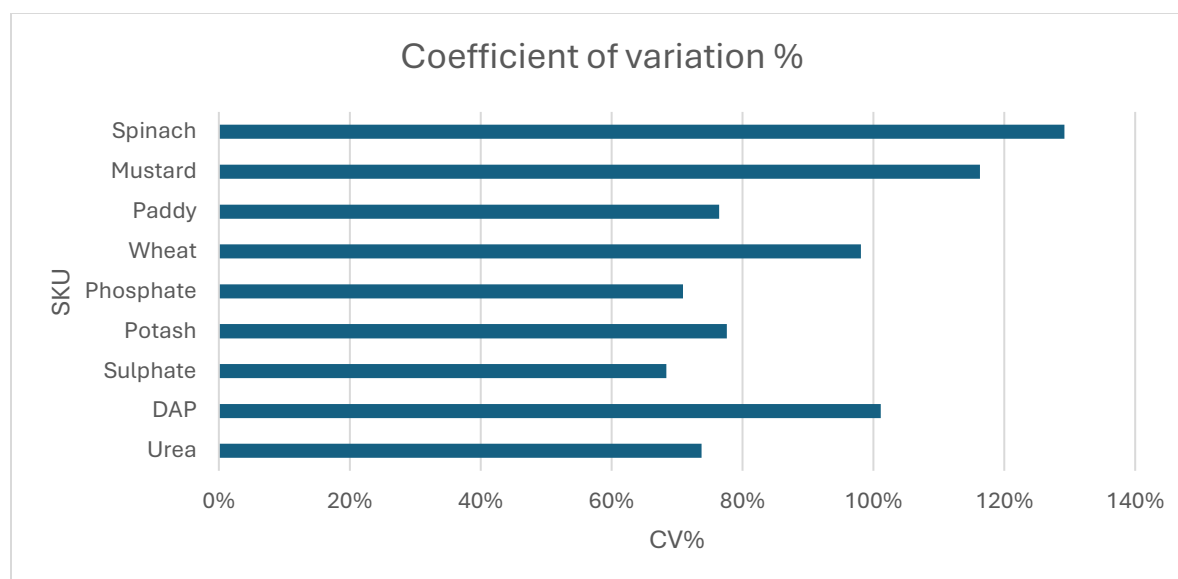


Figure 3: This bar chart showing demand variability

The coefficient of variation analysis evaluates demand stability for fertilizers, seeds, and crops by comparing standard deviation relative to average demand. The CV chart indicates that most SKUs experience moderate to high demand variability, which directly affects inventory risk and safety stock requirements. Spinach shows the highest variability with a CV of 129%, followed by Mustard at 116% and DAP at 101%. These items exhibit highly unpredictable demand patterns and therefore require higher safety stock levels to prevent stockouts during sudden demand surges. Wheat also demonstrates significant variability (98%), indicating fluctuating seasonal demand associated with crop cycles. In contrast, Sulphate (68%), Phosphate (71%), and Urea (74%) display relatively lower variability, suggesting more stable consumption patterns. These products can be managed using more consistent replenishment policies with comparatively lower buffer inventory. Paddy, despite having the highest mean demand, shows moderate variability (76%), indicating strong but predictable seasonal demand aligned with monsoon sowing periods. From an inventory management perspective, high-CV products increase uncertainty in demand forecasting and require dynamic inventory policies. Maintaining insufficient stock for these items may lead to shortages during peak agricultural seasons, while excessive stocking may increase holding costs.

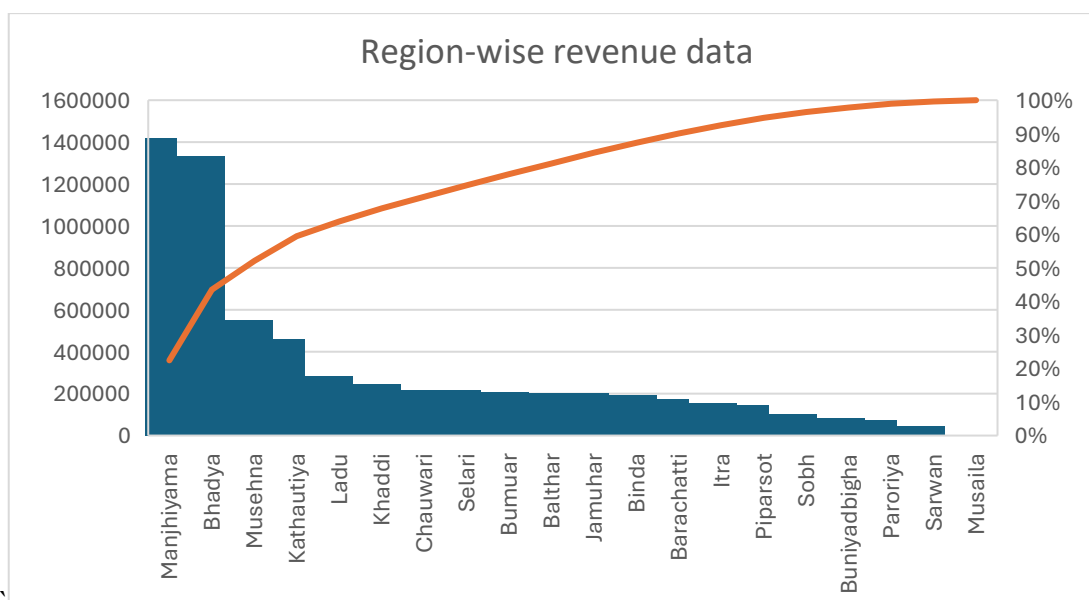


Figure 4: This pareto chart showing total revenue generated by particular region

The objective of regional sales performance analysis is to evaluate how total revenue is distributed across different villages and market area served by **Krishi Khad Bhandar**. This analysis helps identify high performing regions, moderate contributors, and underperforming areas, enabling region specific sales, inventory and distribution strategies.

- **Manjhiyama** and **Bhadya** emerge as the **top-performing regions**, contributing the highest share of total revenue ₹1421605 and ₹1336140 respectively. These regions significantly outperform others, indicating, strong and consistent demand, higher concentration of farmers and dealers, better market penetration and customer loyalty. These region as core revenue centers and critical to the overall business performance. High performing regions maintain high inventory availability, offer premium services, strengthen dealer relationships.
- Region such as **Kathautiya**, **Musehna**, **Ladu**, **Khaddi** show moderate revenue contributions ₹462026, ₹549847, ₹282820, ₹246519 respectively. These areas demonstrate stable demand but not reach peak performance levels. These regions have growth potential and can be improved through targeted promotions, better product availability during peak seasons, dealer incentives and credit facilities.
- Regions including **Buniyadbigha**, **Paroriya**, **Piparsot**, **Sarwan**, **Musaila**, **Sobh** contribute relatively low revenue ₹84802, ₹76175, ₹144170, ₹43840, ₹19670, ₹104470 respectively. Low performance may be due to smaller farmer base,

recognition easier than numeric tables. Here's a breakdown of key observations based on the correlation coefficients:

High Correlations:

- **Potash and Urea (0.96148):** A strong correlation indicates that the sales of Potash and Urea often move together. These products have shared demand due to specific nutrients and much lower price than DAP. Farmers often purchase them in the same cropping cycle, leading to simultaneous demand and high co-movement in sales.
- **Sulphate and DAP (0.88941311):** A significant strong correlation between these fertilizer indicates that much essential fertilizers among farmer despite of higher prices. DAP supplies phosphorus, while Sulphate provides sulphur, both essential during early root and shoot development. These fertilizers are commonly used together at the basal stage, especially in cereal and pulse crops, resulting in complementary purchasing behavior.
- **Potash-Sulphate (0.88392) and Urea - Sulphate (0.88045):** Another strong correlation between these fertilizers reinforces their complementary demand. Suggest bundled demand or similar seasonal usage. Potash and Urea are frequently combined with Sulphate to improve nutrient efficiency and crop quality. Their demand peaks during the same seasons, and farmers often buy them as a nutrient package, creating strong positive correlations.

Moderate Correlations:

- **Phosphate - Potash (0.728844) and Phosphate - Urea (r = 0.702584):** These nutrients are part of balanced fertilization, but their application depends on soil nutrient deficiencies and crop requirements. Farmers may already have adequate phosphorus in the soil, so phosphate is not applied every season, reducing perfect co-movement with Potash or Urea.
- **DAP - Urea (0.71) and DAP - Potash (0.67):** DAP is mainly used as a **basal fertilizer**, while Urea and Potash are often applied in split or later stages of crop growth. This difference in timing and quantity of application weakens the strength of the relationship, making it moderate rather than strong.
- **Sulphate - Zyme (0.48):** Sulphate is a core nutrient fertilizer, whereas Zyme is a growth promoter/micronutrient supplement. Zyme is applied selectively based on

crop stress or farmer preference, so it is only partially complementary to Sulphate, resulting in a moderate correlation.

Low and Negligible Correlations:

- **Zyme with Most Products ($r \approx 0.01-0.23$):** The consistently low correlation of zyme with other products suggests that it is purchased independently of other items and doesn't show a strong relationship with sales patterns. Its purchase is not mandatory in every cropping cycle and is often decided independently of major fertilizer requirements, resulting in very weak or negligible correlation.
- **DAP – Pest Control (0.03):** DAP (a fertilizer) is applied during nutrient management stages, while pest control products are used only when pest infestation occurs. Since these two inputs serve completely different purposes and are driven by different timing and triggers, their sales show almost no relationship.

Interpretation of Results and Recommendations :

Problem 1: Challenges in Understanding Sales Trends Over Time

Interpretation:

- From figure 1 of seeds data, major peaks are observed during crop-specific sowing periods. In 2024, paddy and wheat seeds show extremely high sales during July and November respectively, reflecting Kharif and Rabi seasons demand. Similarly, in 2025, paddy sales peak sharply in July, while wheat shows higher sales toward the end of the year. Other seeds such as maize, mustard, and moong display moderate seasonal demand aligned with regional cropping patterns.
- From figure 2 of fertilizers data, it follow a similar seasonal structure. Products such as urea and DAP show strong growth from June to November, corresponding to land preparation and crop nutrition requirements. November 2024 records the highest fertilizer revenue, indicating heavy Rabi season stocking. However, in 2025, sales appear more evenly distributed, suggesting improved inventory planning or changes in demand patterns.
- The primary challenge in understanding sales trends over time arises from high seasonality causing irregular revenue patterns, extreme peak months that distort

average monthly sales, multiple product categories with different seasonal cycles, difficulty in distinguishing growth trends from seasonal fluctuations.

Recommendations:

- I. Implement Seasonal Decomposition Analysis: Use time series methods such as seasonal Index or moving average to separate like trend component which shows long term pattern; the seasonal component, which reflects recurring variations over time.
- II. Develop monthly sales forecasting model: To develop a monthly sales forecasting model, you can use seasonal moving average. It is a simpler method that factors in seasonality by averaging sales over specific periods.
- III. Improve Inventory Planning: Owner can increase stock levels before June-July (Kharif season), increase stock levels before October-November (Rabi season) and reduce excess stock during March-May (lean Period)
- IV. Create Product-Wise Seasonal Strategy: Since seeds and fertilizers have different peak periods: Plan marketing campaigns before sowing seasons, introduce early booking discounts for farmers.
- V. Use Data visualization Dashboards: Develop a power BI or tableau dashboard to monitor monthly trends, compare year on year growth and Identifying high-performing products.

Problem 2: Inventory Management Issues

Interpretation:

- The inventory analysis reveals that demand for seeds and fertilizers is highly seasonal and uneven, leading to two major operational problems: stockouts during peak farming months and excess inventory during off-seasons periods.
- From table 2, the coefficient of variation (CV) shows significant demand uncertainty; Very high variability: Spinach(129%), mustard(116%); moderate variability: urea(74%), paddy(76%); relatively stable: Sulphate(68%), phosphate(71%). This means demand does not follow a smooth monthly pattern. Instead, purchases are concentrated around sowing seasons, which explains the large spikes.
- Urea shows consistent seasonal demand. It records moderate but steady sales throughout the year. Demand increases during August and November-December, following seed sowing activity. This indicates Urea's role as a basic and essential input.
- Fertilizer demand is seasonal but relatively more stable than seeds. Pesticides show the highest variability, indicating irregular demand. High variability

increases risk of stockouts during peak season. Off-season stocking leads to overstocking and capital blockage. Lack of real time inventory tracking worsens inefficiencies.

- From the combined analysis, the core problem are acquisition based on average demand, no differentiation between stable and volatile products, lack of demand-driven reorder policy.

Recommendations:

- I. Seasonal Inventory Planning: Procurement should follow agricultural cycles such as pre-stock high variability seeds and fertilizers before sowing months, reduce purchasing during off-season periods, align supply planning with regional crops patterns.
- II. CV-based Inventory policy: Adopt differentiated control strategies, if $CV > 100\%$ then safety stock should be high and frequently review the stock, if CV is in between 70-100% then safety stock should be moderate and seasonal ordering, if $CV < 70\%$ then it can be standard replenishment policy.
- III. Use sales forecasting to adjust inventory levels during high-demand months, particularly for fast moving and time-sensitive fertilizers such as **DAP** and **Urea**.
- IV. Maintain higher safety stock during peak seasons only; products with **high Coefficient of Variation (CV)** should follow aggressive seasonal stocking, while products with **low CV** should maintain a moderate buffer stock.
- V. Bundle-Based Stocking Strategy: From table 3 , we can analyze that Potash, Urea, Sulphate, DAP, and Phosphate move together: Plan joint procurement, maintain similar service levels use combined demand. This reduces lost sales due to incomplete nutrient availability.

Problem 3: Challenges in Region-Wise Revenue Distribution

Interpretation:

- From figure 4. the analysis shows that a large portion of total revenue comes from only a few regions, indicating uneven regional performance. **Manjhiyama** and **Bhadya** are the highest-performing regions and together contribute a major share of sales. These regions benefit from higher farmer population, better road connectivity, and easier access to agricultural inputs. Medium-performing regions show consistent but moderate sales, suggesting that demand exists but is not fully developed.
- Revenue concentration is visible in the top-performing regions, which dominate total income. This suggests that customer demand, supply access, or product

availability is significantly stronger in these areas compared to others. Such dependency increases business risk because performance becomes sensitive to disruptions in only a few markets.

- Several regions exhibit consistently low revenue contribution. This does not necessarily indicate low demand; instead, it may reflect distribution inefficiencies, lack of awareness, or weak inventory availability. In rural agricultural supply chains, demand is often present but unmet due to logistics and stocking gaps.
- Several regions contribute very low revenue, possibly due to fewer farmers, poor irrigation facilities, or limited awareness. Heavy dependence on a small number of regions increases business risk, especially if demand falls due to weather or crop failure. Regional variations in sales are influenced by differences in irrigation availability, crop types grown, purchasing capacity, and distance from the store.

Recommendations:

- I. High-performing regions should be given priority in inventory stocking to avoid shortages during peak demand periods. The business should strengthen relationships with farmers and dealers in top regions through better service and timely supply. Medium-performing regions can be improved by introducing targeted promotional offers and awareness campaigns.
- II. Region-Based Inventory Allocation: Stock levels should be planned according to regional demand intensity rather than uniform distribution. High-performing regions require consistent availability, while emerging regions need targeted supply expansion.
- III. Demand Forecasting by Region: Use historical sales and seasonal patterns to forecast region-specific demand. This aligns procurement with local crop cycles and reduces both stockouts and excess inventory.
- IV. Expansion Strategy for Low-Revenue Regions: Instead of treating low revenue as low demand, investigate; supply accessibility, retailer presence awareness among farmers, logistics delays, targeted distribution improvement can unlock hidden demand.
- V. Risk Diversification Strategy: Reducing dependency on a few regions stabilizes revenue flow. Balanced geographic coverage improves long-term sustainability and reduces vulnerability to regional disruptions.

Problem 4: Space Constraint

Interpretation:

- During pre-season stocking: Large quantities of high-demand fertilizers arrive together. Correlated products occupy storage simultaneously. Warehouse utilization approaches maximum capacity. During off-season: Slow-moving items occupy shelf space. Fast-moving SKUs may still require replenishment. Storage becomes inefficiently allocated.
- High-CV Products Increase Storage Risk : Products with high variability require safety stock buffers. However, storing excessive safety stock for volatile items leads to overstocking risk, increased holding cost, expiry or deterioration (especially for seeds) blocked space for faster-moving SKUs. Space constraint is therefore directly linked to poor demand forecasting and safety stock miscalculation.
- Correlated Products Amplify Space Pressure: Because Potash, Urea, Sulphate, and DAP move together, they are stocked together. When demand rises: All these SKUs require replenishment at once. combined volume creates storage bottlenecks. This explains why space shortages occur seasonally rather than consistently.

Recommendations:

- I. CV-Based Dynamic Stocking: High CV items review frequently and controlled safety stock. Moderate CV should plan seasonally and low CV should stable replenishment avoid excessive buffer stock for highly volatile SKUs.
- II. Pre-Season Staggered Procurement: Instead of bulk purchasing at once owner should split procurement into phased deliveries and use rolling replenishment before peak months and reduce one-time storage overload.
- III. Cross-Docking Strategy: For high-demand fertilizers, owner should directly dispatch from supplier to regional outlets and minimize warehouse holding time. This reduces storage pressure.
- IV. Data-Driven Space Optimization: Owner should use forecasting technique for peak storage and recommend optimal order quantity based on available space and alert when projected inventory exceeds capacity.